

SIMPLE LINEAR REGRESSION

Dr Sandheep Sugathan
MBBS MD DPH FRSPH FAIMER Fellow

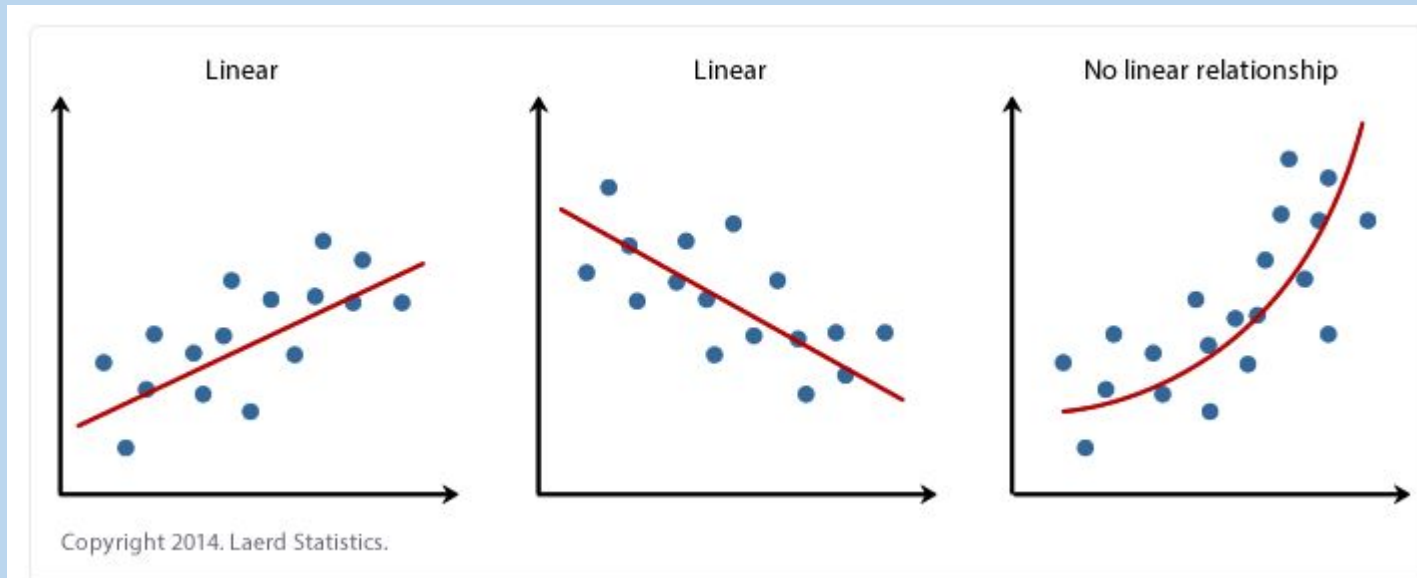
A simple linear regression assesses the linear relationship between **two continuous variables** and to **predict the value of a dependent variable** based on the value of an independent variable.

it will let you:

- (a) determine whether the linear regression between these two variables is statistically significant;**
- (b) determine how much of the variation in the dependent variable is explained by the changes in independent variable;**
- (c) understand the direction and magnitude of linear relationship; and**
- (d) predict values of the dependent variables based on different values of the independent variable.**

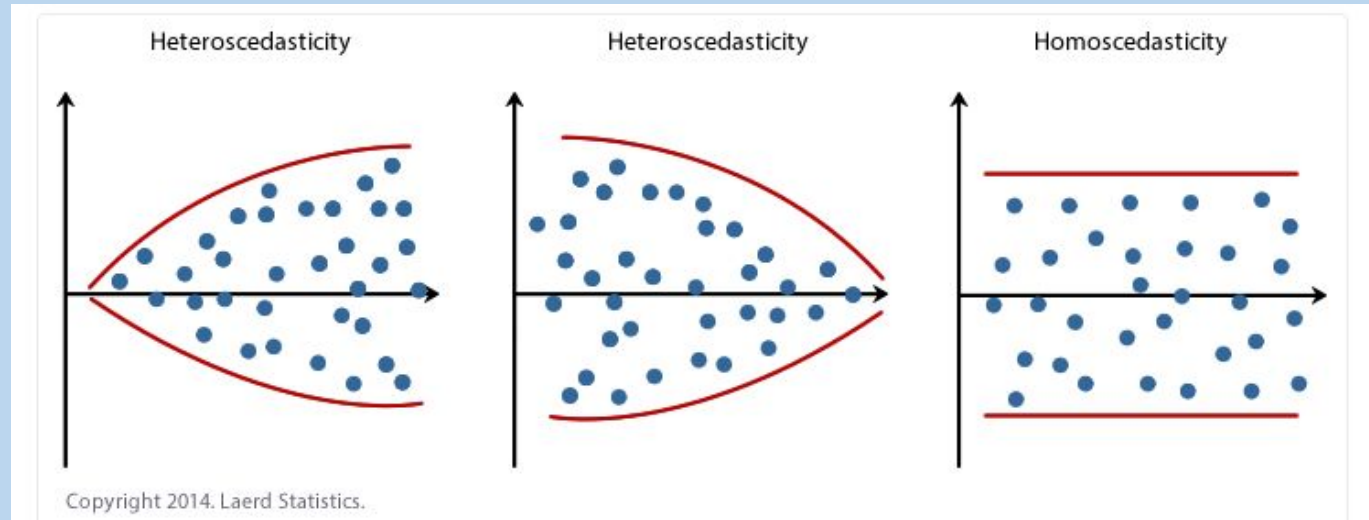
6 assumptions must be satisfied for doing a linear regression analysis

1. you have a continuous dependent variable;
2. you have a continuous independent variable.
3. there is a linear relationship between the two variables;

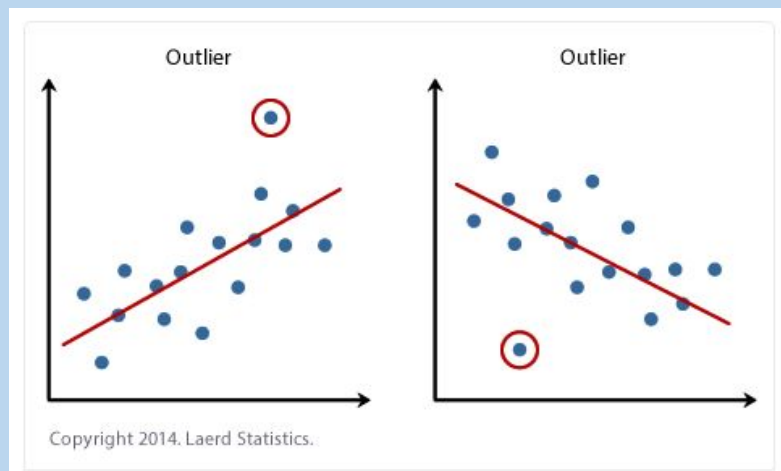


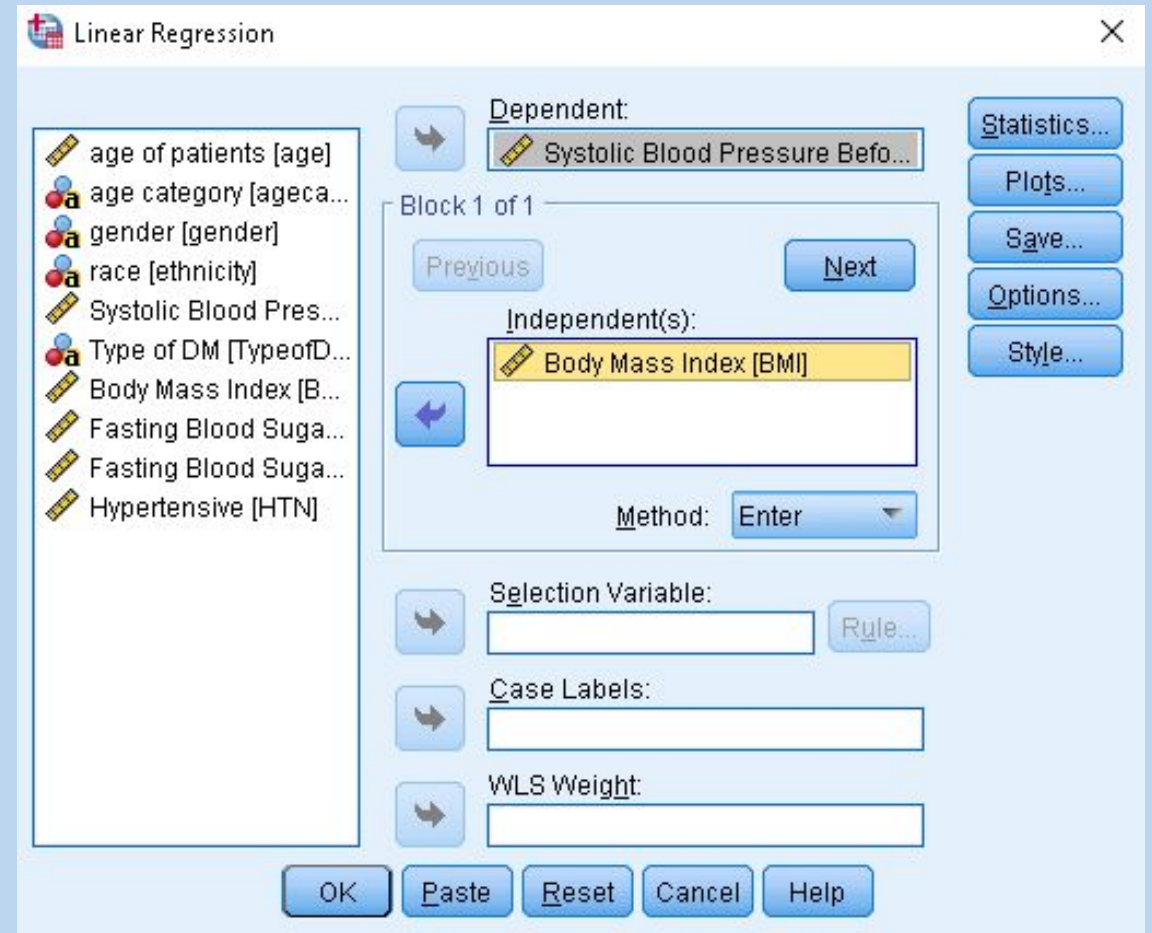
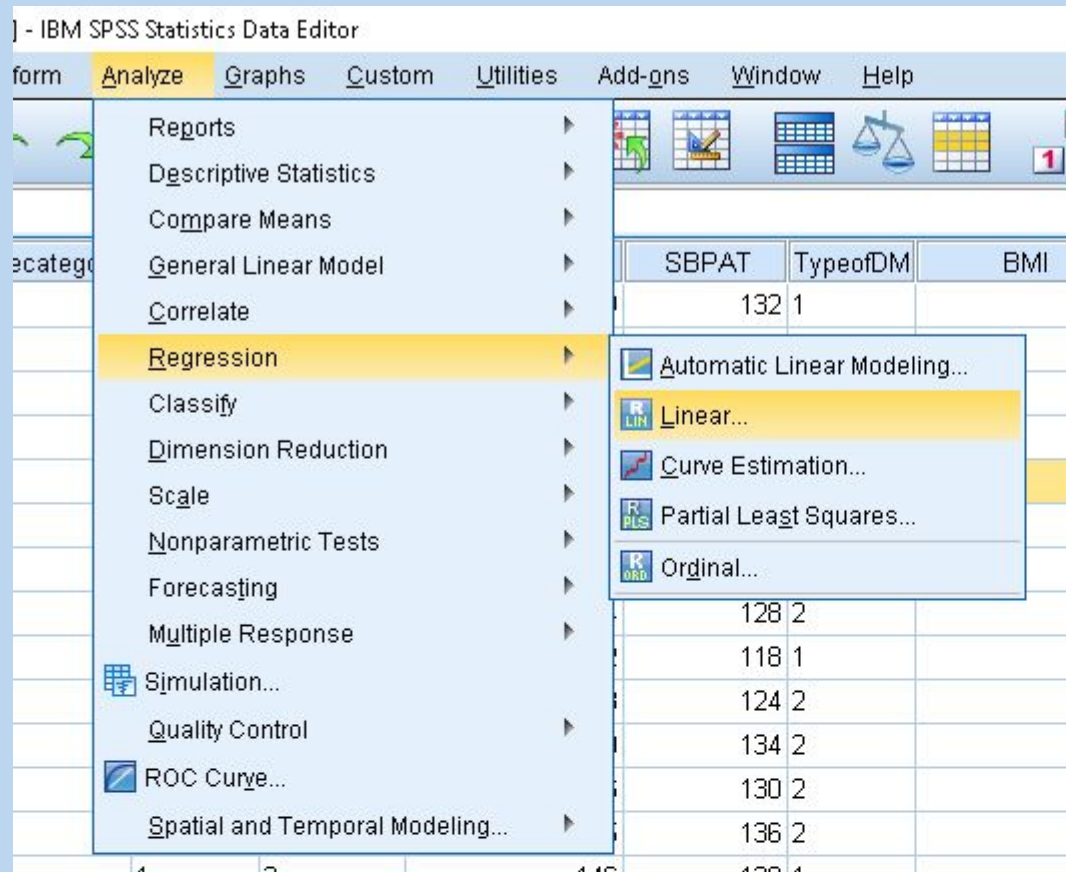
4. you have independence of observations; - Durbin Watson Statistics should be near to 2 – shows non-autocorrelation

5. there is homoscedasticity;



6. there are no significant outliers





Click Statistics

Linear Regression: Statistics

Regression Coefficients

- ☒ Estimates
- ☐ Confidence intervals
Level(%): 95
- ☐ Covariance matrix

Model fit

- ☒ Model fit
- ☐ R squared change
- ☐ Descriptives
- ☐ Part and partial correlations
- ☐ Collinearity diagnostics

Residuals

- ☐ Durbin-Watson
- ☐ Casewise diagnostics
 - ☐ Outliers outside: 3 standard deviations
 - ☐ All cases

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Click continue

Results

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.884 ^a	.781	.775	3.190	1.225

a. Predictors: (Constant), Body Mass Index

b. Dependent Variable: Systolic Blood Pressure Before Treatment

R is the correlation coefficient – shows strong positive correlation

R squared shows - represents the percentage of variance in the dependent variable (SBP) that can be explained by the independent variable (BMI). - here it is 78.1%

The adjusted *R square* value corrects positive bias to provide a value that would be expected *in the population*. - 77.5%

ANOVA shows statistical significance of the model

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1378.342	1	1378.342	135.470	.000 ^b
	Residual	386.633	38	10.175		
	Total	1764.975	39			

a. Dependent Variable: Systolic Blood Pressure Before Treatment

b. Predictors: (Constant), Body Mass Index

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	.230	11.867		.019	.985	-23.793	24.253
	Body Mass Index	5.190	.446	.884	11.639	.000	4.287	6.092

a. Dependent Variable: Systolic Blood Pressure Before Treatment

Regression coefficient for BMI is 5.19 (95% CI is from 4.287 to 6.092). It is statistically significant (P value less than 0.001)

$$\text{SBP} = 0.230 + 5.190 * \text{BMI value}$$